

whole.

The [Reserved Bit Fields](#) conventions define reserved bitmap data.

- It is RECOMMENDED to define more bits than initially needed to be able to support more values for later revisions.
- The Bitmap type MAY be used to support up to 8, 16, 32 or 64 boolean values.
- Bits MAY be combined to enumerate other values.
- Bits SHOULD be combined as contiguous bit fields.
- Future revisions MAY require non-contiguous bit fields.
- The conformance for a bit in a bitmap SHALL be mandatory or dependent upon an existing discoverable element, and therefore SHALL NOT be purely optional.

Allowable Conformance for a bit in a bitmap:

- Mandatory
- Dependent upon a Feature supported in the FeatureMap attribute.
- Dependent upon the support of an attribute.

A nullable bitmap SHALL NOT permit use of the most significant bit.

7.19.1.3. Unsigned Integer (8, 16, 24, 32, 40, 48, 56 and 64-bit)

This type represents an unsigned integer with length of N bits and a usable range of:

- $[0..2^N-1]$ if not nullable OR
- $[0..2^N-2]$ if nullable.

The following table presents the representable values following the above rules:

Width N (bits)	Minimum value	Maximum value if nullable	Maximum value if not nullable
8	0 (0x00)	254 (0xFE)	255 (0xFF)
16	0 (0x0000)	65534 (0xFFFE)	65535 (0xFFFF)
24	0 (0x000000)	16777214 (0xFFFFFE)	16777215 (0FFFFFFF)
32	0 (0x00000000)	4294967294 (0xFFFFFFFEE)	4294967295 (0xFFFFFFFF)
40	0 (0x0000000000)	1099511627774 (0xFFFFFFFFFFE)	1099511627775 (0xFFFFFFFFFFF)
48	0 (0x000000000000)	281474976710654 (0xFFFFFFFFFFFEE)	281474976710655 (0xFFFFFFFFFFFFF)